

Deep Learning 2025 – Project Report

Project Title: GAN CNN Hybrid Model for Image Classification

Course: Deep Learning 2025

Instructors: Golnaz Sahebi, Pertti Ranttila

Group Members: Imad, Shrish, Ritesh, Vlad

1. Division of Tasks

The project was divided into four milestones. Each pair of students was responsible for two milestones, which ensured balanced workload and clear responsibilities.

Milestone 1: Dataset Preparation and Environment Setup

Responsible: Imad and Shrish

Main tasks: Preparing and organizing the dataset, cleaning and resizing images, setting up the development environment, installing dependencies, and preparing the initial project documentation.

Milestone 2: Baseline CNN Model

Responsible: Imad and Shrish

Main tasks: Implementing the baseline CNN classifier, running early training experiments, applying basic regularization methods, tuning initial parameters, and generating preliminary results.

Milestone 3: GAN Integration and Model Improvement

Responsible: Ritesh and Vlad

Main tasks: Adding the GAN model for synthetic data generation, improving training stability, enhancing the CNN structure, applying techniques to handle class imbalance, and optimizing the training process.

Milestone 4: Final Evaluation and Documentation

Responsible: Ritesh and Vlad

Main tasks: Running final training cycles, evaluating accuracy and other metrics, saving

the completed model, finalizing documentation, and preparing the project for submission.

2. Active and Passive Group Members

All four members contributed actively to their assigned milestones.

Imad: Strong contribution to early project setup and CNN implementation.

Shrish: Actively collaborated on the baseline model and tuning tasks.

Ritesh: Major contribution to GAN development and model optimization.

Vlad: Significant involvement in training, monitoring results, and preparing the final evaluation material.

There were no passive members. Everyone completed their responsibilities and took part in discussions and problem solving.

3. Challenges

The group encountered several technical and practical challenges:

1. Dataset imbalance created biased predictions and required additional solutions such as GAN-generated samples and class weights.
2. GAN training instability caused mode collapse and required several rounds of tuning.
3. Limited computational resources increased training time and slowed experimentation.
4. Scheduling and coordinating training sessions required consistent communication.
5. Hyperparameter tuning took more time than expected due to the complexity of both models.

4. Reflections on Teamwork

The project benefited from clear communication and a well-structured division of work. Each member focused on their milestone but also supported the others when needed.

Discussions helped improve the models, solve technical issues, and align the different parts of the project.

Team members learned from each other, especially when reviewing the CNN and GAN components. The group collaborated effectively, maintained a positive working atmosphere, and successfully completed the project with shared effort and commitment.